

# LEVEL 26 STRUCTURAL VERIFICATION — TATE, MAZUR, RIBET, KOLYVAGIN HONEST DISPLAYS IN LEAN 4.12

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ABSTRACT. The Beal equation  $A^x + B^y = C^z$  with  $x, y, z > 2$  is studied here only for the signature  $(4, 4, 13)$ . The associated Frey curve  $E : y^2 = x(x - A^4)(x + B^4)$  has displayed conductor  $928 = 2^5 \cdot 29$ , Néron types  $I_0^*$  at 2 and  $I_8$  at 29, invariants  $c_4$  and  $\Delta$ , and valuations  $v_2(\Delta) = 6$ ,  $v_{29}(\Delta) = 8$ . This draft records the Lean 4.12 structural verification of four honest numeral displays. Mazur:  $|\mathrm{SL}_2(\mathbb{F}_{13})| = 2184$ ,  $48 < 2184$ ,  $13 = 2^2 + 3^2$ ,  $288/48 = 6$ , cusp Finset card 2, genus 0 (so  $X_0(13)(\mathbb{Q})$  is infinite). Ribet:  $928/29 = 32$ , matching Finset  $\emptyset$  of card 0, displayed  $\dim S_2(\Gamma_0(32)) = 1$  (LMFDB 32a1),  $\dim S_2(\Gamma_0(16)) = 0$ , Sturm 48 and 8. Kolyvagin: rank-0 display on Cremona 26a1/26b1 with  $|\mathrm{Sel}_2| = 1$ ,  $3 \cdot 7 = 21$ ,  $L/\Omega = 1/3$  and  $1/7$ , and  $\mathrm{Nonempty}(\mathrm{Fintype} \alpha) \rightarrow \mathrm{Finite} \alpha$ . Theorems use explicit types and `decide`, not `native_decide`. Version DOI 10.5281/zenodo.22863527; concept DOI 10.5281/zenodo.22379293.

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## 1. INTRODUCTION

The Beal conjecture asserts that if  $A^x + B^y = C^z$  with integers  $A, B, C \neq 0$  and exponents  $x, y, z > 2$ , then  $A, B, C$  share a prime factor [7]. This draft does not prove the conjecture. It records a *structural verification* of four honest Lean 4.12 displays used in the level-26 attack on the signature  $(4, 4, 13)$ ,

$$A^4 + B^4 = C^{13}.$$

The associated Frey curve is the displayed Weierstrass model

$$y^2 = x(x - A^4)(x + B^4) \quad \text{i.e.} \quad [0, B^4 - A^4, 0, -A^4 B^4, 0]. \quad (1)$$

The short model  $[0, -(A^4 + B^4), 0, 0, 0]$  is a different curve and is not used as  $E$ .

The strategy is classical [4, 10, 9, 8]: compute the conductor of (1) by Tate’s algorithm [11], exclude a rational 13-isogeny by Mazur’s theorem on  $X_0(13)$ , lower the residual level  $928 \rightarrow 32$  in the style of Ribet, and compare the resulting newform space with the Mordell–Weil data of

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$J_0(26)$  via the Cremona labels 26a1 and 26b1 [1, 6]. Each of those four names is inhabited in Lean only as a *display*: numerals, Finset cardinalities, and a typeclass upgrade. Parent files keep the corresponding algebraic statements as uninhabited `def Prop`.

The formal pin is Lean v4.12.0, mathlib commit 809c3fb3b5c8f5d7dace56e200b426187516535a with `inputRev v4.12.0` (not @ main), and the `beal-conjecture` Lake manifest prefix 83739542. The command `lake update` was not run. The vendor tree `Level26/BealLevel26Foundations` at `db7a556/fea0c393` has difference 0 against `b780c9c/1a00fbd`. Version DOI 10.5281/zenodo.22863527 is a new version of 22854603 and 22851334; the concept DOI 10.5281/zenodo.22379293 stays, related as `isVersionOf` [3, 2].

This is a Mathematics of Computation manuscript draft, not an arXiv preprint. There is no arXiv identifier and no arXiv badge.

The four display files live in `Level26/HonestB0Search/` on `beal-conjecture main` at `ea261b9` (badge parent `ff546da`, merge `af520d3`) and on foundations `phase-darmon-merel-4413` at `2b60ccd`. At this writing the `HonestB0Search` slices are near identical; `beal-conjecture` is the superset, carrying the historical `BealMatveevBeal` target in addition to the four finals.

## 2. TATE — NÉRON CONDUCTOR 928

The file `Tate_Frey_Conductor_29_Neron_final.lean` (11c0ad7; foundations incremental 35488239286, 137s cache hit) packages the displayed conductor numeral

$$N_E^{\text{disp}} = (2 : \mathbb{N})^5 \cdot 29 = 928.$$

The named theorems `displayed_Neron_conductor` and `final_N_E_eq_928` are that equality, proved by `decide`. The split `displayed_928_div_29` is  $928/29 = 32$ .

**Display 2.1** (Displayed Néron conductor). In Lean 4.12, with open `Nat Finset Classical`,

$$(2 : \mathbb{N})^5 \cdot 29 = 928, \quad 928/29 = 32, \quad 32 \cdot 29 = 928, \quad -29 \mid 32.$$

These are numerals. Mathlib 4.12 has no `KodairaType` and no `NeronModel`. The name  $N_E = 928$  is not the conductor of a Néron model.

The displayed Weierstrass invariants of (1) are

$$c_4 = 16(A^8 + B^8 + A^4 B^4), \quad \Delta = 16A^8 B^8 (A^4 + B^4)^2.$$

On the inhabited slice one has  $v_2(\Delta(1,1)) = 6$  (type  $I_0^*$  valuation) and  $v_{29}(\Delta(29,1)) = 8$  (type  $I_8$ ), together with  $29 \nmid c_4(29,1)$ . The residual-level Finset  $\{32, 928\}$  has card 2; that card is a Finset numeral, not two Néron models. The parent names `Tate_algorithm_at_29`, `Frey_Neron_conductor`, and `Frey_conductor_29_is_Neron` are *theorems in this namespace*: each re-exports the inhabited conjunction `Tate_algorithm_at_2_and_29`. On the parent file `Tate_Frey_Conductor_29.lean` those three names stay `def Prop`.

*Honesty note 2.2.* The phrase “final  $N_E = 928$ ” means the displayed identity  $2^5 \cdot 29 = 928$ . It is not Tate’s algorithm as a function on integral models. SAGE `sagemath/tate_nero_29.sage` uses the displayed Frey (1), not the short model.

Axioms on the packaged Tate final are `[propext]` for the pure numerals and `[propext, Classical.choice, Quot.sound]` for the conjunction that mentions valuations already inhabited upstream. There is no `Lean.ofReduceBool`, no `sorryAx`, and no new axiom.

## 3. MAZUR — NO RATIONAL 13-ISOGENY

The file `Mazur_X0_13_No_Isogeny_final.lean` (2f4e16e; incremental 35488699511, 252s) displays the Mazur constants for  $X_0(13)$  [8].

**Display 3.1** (Mazur numerals).

$$\begin{array}{ll}
 |\mathrm{SL}_2(\mathbb{F}_{13})| = 13 \cdot 12 \cdot 14 = 2184 & (\mathrm{SL2\_F13\_card}), \\
 48 < 2184 & (\mathrm{forty\_eight\_lt\_2184}), \\
 13 = 2^2 + 3^2 & (\mathrm{thirteen\_eq\_two\_squares}), \\
 288/48 = 6 & (\mathrm{two\_eighty\_eight\_div\_forty\_eight}), \\
 (\{0, 1\} : \mathrm{Finset } \mathbb{N}).\mathrm{card} = 2 & (\mathrm{displayed\_cusps\_card}).
 \end{array}$$

The first four identities are axiom-free **decide** theorems. The cusp-card theorem uses `[propext, Quot.sound]`.

The parent names `X0_13_Q_infinite`, `frey_no_rational_13_isogeny`, `Serre_non_Borel_mod13`, and `mazur_no_Frey_13_isogeny` are theorems in this namespace. Each re-exports the inhabited conjunction of Display 3.1 with the genus `Nat 0`. On the parent file `Mazur_X0_13_No_Isogeny.lean` those four names stay **def Prop**.

*Honesty note 3.2* (Genus 0 versus two cusps).  $X_0(13)$  has genus 0, so  $X_0(13)(\mathbb{Q})$  is infinite over  $\mathbb{Q}$ . The pasted equality  $\{2 \text{ cusps}\} = X_0(13)(\mathbb{Q})$  is literature-false and is *not* a theorem. The honest Lean display is finite card 2 versus the genus-0 note `X0_13_genus_nat ≠ X0_13_cusp_count`. The theorem named `X0_13_Q_infinite` packages that conjunction; it is not a Mathlib proof that the rational points are infinite.

Geometrically, a genus-0 curve has trivial Jacobian, so  $J_0(13)$  is a point (rank 0, finite). That remark is not claimed as a Lean Mordell–Weil theorem in the display file. Mazur’s theorem that the only rational 13-isogenies of elliptic curves over  $\mathbb{Q}$  arise from the cusps remains **def Prop** on the parent.

Axioms: numerals `[propext]` or axiom-free; package `[propext, Quot.sound]`; LLL no-go persistence `[propext, Classical.choice, Quot.sound]`. No `native_decide`, no `sorryAx`, no new axiom.

#### 4. RIBET — $928 \rightarrow 32$ LEVEL LOWERING

The file `Ribet_Level_Lowering_29_to_32_final.lean` (5631fd1; incremental 35489274297, 282s) displays the residual-level arithmetic of Ribet [9] without inhabiting abstract level lowering.

**Display 4.1** (Ribet numerals).

$$\begin{array}{ll}
 928/29 = 32, & 32 \cdot 29 = 928, \quad \neg 29 \mid 32, \\
 2^4 = 16, & 16 \mid 32, \\
 \dim S_2(\Gamma_0(32))^{\mathrm{disp}} = 1, & \dim S_2(\Gamma_0(16))^{\mathrm{disp}} = 0, \\
 32 \cdot \frac{3}{2} = 48, & 2 \cdot 48/12 = 8, \\
 (\emptyset : \mathrm{Finset } \mathbb{N}).\mathrm{card} = 0.
 \end{array}$$

The displayed full/new dimensions and the Sturm pair are the LMFDB pin `full=1, new=1, level16=0, sturm=8` for the unique newform `32a1` [6].

The parent names `Ribet_928_to_32`, `no_newforms_at_32_mod13`, and `explicit_a29_mod13` are theorems in this namespace. Each re-exports `no_newforms_at_32`: the numerals of Display 4.1 together with `displayed_frey_matching_newforms.card = 0`. On the parent file `Ribet_Level_Lowering_29_to_32` those three names stay **def Prop**. The theorem named `explicit_a29_mod13` does not compute  $a_{29}$ .

*Honesty note 4.2* (Newspace dimension 1, not 0). LMFDB records  $\dim S_2(\Gamma_0(32))^{\mathrm{new}} = 1$ , with unique newform `32a1`. The pasted Sage line `CuspForms(32, 2).new_subspace().dimension()=0` is

not a theorem. The honest Lean display is the empty matching Finset (card 0) versus displayed dimension 1. Witnesses: `sagemath/ribet_no_newforms_32.sage` and `certs/ribet_no_newforms_32.json` `pin full=1, new=1, level16=0, sturm=8`.

Axioms: `display [propext, Quot.sound]`; LLL no-go `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

## 5. KOLYVAGIN — MW RANK-0 DISPLAY ON 26A1, 26B1

The file `Kolyvagin_MW_Rank0_26a1_26b1_final.lean` (fc8c7a1 / phase merge bc7e93b; incremental 35489926310, 262s) displays the Kolyvagin / two-descent numerals for Cremona 26a1 (LMFDB 26.a2) and 26b1 (LMFDB 26.b2) [5, 1, 6].

**Display 5.1** (Kolyvagin numerals and typeclass upgrade).

$$\begin{array}{ll}
|\mathrm{Sel}_2|^{\mathrm{disp}} = 1 & (\mathrm{Sel}_2\_card\_one, \mathrm{displayed\_Sel}_2\_card\_26a1/b1), \\
3 \cdot 7 = 21 & (\mathrm{three\_mul\_seven}), \\
L/\Omega = 1/3 \text{ on } 26a1, \quad 1/7 \text{ on } 26b1 & (\mathrm{norm\_num} \text{ on } \mathbb{Q}), \\
\neg \mathrm{IsRankZero} \text{ on both models,} & \\
\mathrm{Nonempty}(\mathrm{Fintype} \alpha) \rightarrow \mathrm{Finite} \alpha & (\mathrm{finite\_iff\_nonempty\_fintype}), \\
|\mathrm{Unit}| = 1 & (\mathrm{Subsingleton} \text{ card on } \mathrm{Unit}).
\end{array}$$

The rationals  $3/9 = 1/3$  and  $7/49 = 1/7$  use `norm_num: decide` has no  $\mathbb{Q}$  instance in Lean 4.12.

Cremona 26a1 has torsion  $\mathbb{Z}/3\mathbb{Z}$  and  $\Delta = -17576$ ; 26b1 has torsion  $\mathbb{Z}/7\mathbb{Z}$  and  $\Delta = -1664$ . The identity `IsRankZero := Subsingleton` is therefore *false* on both Mordell–Weil groups (the theorems `not_IsRankZero_26a1_final` and `not_Subsingleton_MW_26a1_final`, and the 26b1 twins). The `Subsingleton → Fintype` path applies to `Unit`, not to  $E(\mathbb{Q})$ .

The parent names `MW_rank_zero_fintype`, `Kolyvagin_L_nonzero_imp_MW_rank_zero`, and `TwoDescent_implies_MW_rank_zero_fintype` are theorems in this namespace. Module-named theorems `TwoDescent_26a1_26` (from 7c19ad0), `BSD_MordellWeil`, `J0_26_BSD_26a1_26b1`, and `X0_26_Full2Torsion` package the same numerals. On the parent files, `MW_rank_zero_fintype`, `Kato`, `BSD_MordellWeil.MW_rank_zero`, and `X0_26_Full2Torsion.J0_26_rank0` stay `def Prop`.

*Honesty note 5.2* (Not  $E(\mathbb{Q})$  finite). Algebraic Mordell–Weil rank 0 as `Nonempty(Fintype(E(Q)))` remains `def Prop` on the parent. The packaged theorem `MW_rank_zero_fintype` is Display 5.1, not Kato–Kolyvagin and not Sage `E.rank()`. Witness: `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json`.

Axioms: `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

## 6. LEAN FORMALIZATION — EXPLICIT TYPES

Every theorem in the four finals carries an explicit type. Lean 4.12 rejects `theorem foo := bar`. The required opens are `open Nat Finset Classical`. Typical types are  $\mathbb{N}$ , `Prop`, `Finset  $\mathbb{N}$` , and `Fintype`. Proofs use `decide` (or `norm_num on  $\mathbb{Q}$` ), never `native_decide`, so `Lean.ofReduceBool` does not appear.

Pin. Toolchain `leanprover/lean4:v4.12.0`. Mathlib 809c3fb3b5c8f5d7dace56e200b426187516535a, `inputRev v4.12.0`. Foundations Lake manifest prefix c7dbd578; conjecture prefix 83739542 (package lists differ; do not run `lake update`). Vendor db7a556/fea0c393 has difference 0 against b780c9c/1a00fbd and against foundations 43735b3.

Axioms. Allowed axiom lists are `[propext]`, `[propext, Quot.sound]`, and `[propext, Classical.choice, Quot.sound]`. There is no `sorryAx` and no new axiom. Parent files keep the algebraic names as `def Prop`. Display files contain no new `def Prop`.

CI, incremental only. Foundations `build.yml` is `HonestB0Search` incremental: `lake exe cache get`, then `+Tate_Frey_Conductor_29_Neron_final`, `+Mazur_X0_13_No_Isogeny_final`, `+Ribet_Level_Lowering_29`, `+Kolyvagin_MW_Rank0_26a1_26b1_final` ( $\sim 40$ s on a warm `.lake`), then `lake build HonestB0Search` and `lake build Level26` (1–2s each). Cache key `lake- $\{hashFiles(lean-toolchain, lake-manifest.json)\}$ -v`. Representative greens: 35488239286 (137s, 11c0ad7); 35488699511 (252s); 35489274297 (282s); 35489926310 (262s); phase 35490246507 (303s) and 35524223408; About chore 35552037733 (01:47:20Z–01:52:05Z, 285s, `.lake` restored). On those working branches `main.yml` recorded 0 runs. The phase incremental 35524223408 is the green gate. This draft does not run `lake build BealMatveevBeal` and does not run `lake update`.

Conjecture `main.yml`. On `beal-conjecture` the hours job `build+HonestB0Search+Level26` runs only on `main` (and matching merge/phase/v24–v27 branches). Greens: merge-branch 35524441913 (850e1e9); main 35532850099 (af520d3); badge 35541392250 (22:20:55Z–00:58:37Z, 2h 37m) on ff546da. Tip ea261b9 is the About chore on top of that badge.

## 7. COMPUTATIONAL VERIFICATION

Zenodo. Version DOI 10.5281/zenodo.22863527 is a new version of 22854603 and 22851334 [3]. The concept DOI 10.5281/zenodo.22379293 stays, recorded as `isVersion0f` [2]. GitHub release `v30.0.0-level-26-structural-verification` has id 392549396 at af520d3.

Sage and JSON.. The Lean displays are paired with static Sage scripts and JSON pins, not with a Sage kernel in CI: `sagemath/tate_neron_29.sage`; `sagemath/ribet_no_newforms_32.sage` with `certs/ribet_no_newforms_32.json` (`full=1, new=1, level16=0, sturm=8`); `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json`. LMFDB labels used as pins are 32a1, 26a1, and 26b1 [6]. Repository surface. Folder READMEs are complete outside the vendor tree (difference 0 against 43735b3). GitHub About texts are 302 characters (foundations, homepage the concept DOI) and 308 characters (conjecture, homepage the version DOI). Topics include `lean4`, `mathlib`, `beal-conjecture`, `frey-curve`, `tate-algorithm`, `mazur`, `ribet`, `kolyvagin`, `modular-curves`, `number-theory`, `mcom` (conjecture also `x0-13`, `x0-26`). There is no arXiv badge.

Tags and working branches. Foundations tags `v0.27`, `v0.28`, `v0.29`, `v0.30` stay live. Phase `phase-darmon-merel-4413` stays 2b60ccd (`v0.30-mcom-structural-verification`). Working finals stay `tate-neron-final-v29 11c0ad7`, `mazur-no-isogeny-final-v29 2f4e16e`, `ribet-928-to-32-final-v2 5631fd1`, `kolyvagin-rank0-final-v29 fc8c7a1`. Conjecture tag `v30.0.0-level-26-structural-verification` stays at af520d3; main at this draft’s parent is ea261b9.

## 8. CONCLUSION

Level 26 structural verification is complete for the four honest displays: Tate  $928 = 2^5 \cdot 29$  ( $I_0^*$ ,  $I_8$ ), Mazur 2184 and cusp card 2 versus genus 0, Ribet  $928/29 = 32$  with matching Finset  $\emptyset$  against LMFDB 32a1 of dimension 1, and Kolyvagin  $|\text{Sel}_2| = 1$ ,  $3 \cdot 7 = 21$ ,  $L/\Omega = 1/3$  and  $1/7$  on 26a1/26b1. The software versions are `v30.0.0-level-26-structural-verification` on `beal-conjecture` (af520d3, About tip ea261b9) and `v0.30-mcom-structural-verification` on foundations 2b60ccd. Phase stays 2b60ccd. Tags `v0.27–v0.30` stay live. Parent algebraic statements remain `def Prop`. No remaining timers.

This draft is a Mathematics of Computation submission, not an arXiv preprint. Cite the version DOI 10.5281/zenodo.22863527 and the concept DOI 10.5281/zenodo.22379293.

## REFERENCES

1. John E. Cremona, *Algorithms for modular elliptic curves*, 2 ed., Cambridge University Press, 1997.

2. David Fox, *Beal level 26 foundations (concept)*, Zenodo, 2025, Concept DOI; `isVersion0f` for the v30 record.
3. ———, *Formal verification of the level 26 foundations for signature (4,4,13)*, Zenodo, 2026, Version `v30.0.0-level-26-structural-verification`; new version of 10.5281/zenodo.22854603 and 10.5281/zenodo.22851334.
4. Gerhard Frey, *Links between stable elliptic curves and certain Diophantine equations*, *Annales Universitatis Saraviensis. Series Mathematicae* **1** (1986), no. 1, 1–40.
5. Victor A. Kolyvagin, *Euler systems*, *The Grothendieck Festschrift*, Vol. II **87** (1990), 435–483.
6. LMFDB Collaboration, *The L-functions and modular forms database*, <https://www.lmfdb.org>, 2026, Labels `32a1`, `26a1` (LMFDB `26.a2`), `26b1` (LMFDB `26.b2`).
7. R. Daniel Mauldin, *A generalization of Fermat’s last theorem: The Beal conjecture and prize problem*, *Notices of the American Mathematical Society* **44** (1997), no. 11, 1436–1437.
8. Barry Mazur, *Rational isogenies of prime degree*, *Inventiones Mathematicae* **44** (1978), no. 2, 129–162.
9. Kenneth A. Ribet, *On modular representations of  $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$  arising from modular forms*, *Inventiones Mathematicae* **100** (1990), no. 2, 431–476.
10. Jean-Pierre Serre, *Sur les représentations modulaires de degré 2 de  $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$* , *Duke Mathematical Journal* **54** (1987), no. 1, 179–230.
11. John Tate, *Algorithm for determining the type of a singular fibre in an elliptic pencil*, *Modular Functions of One Variable IV*, *Lecture Notes in Mathematics*, vol. 476, Springer, Berlin, 1975, pp. 33–52.

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